崑山科技大學日間部資訊工程系四技智慧機械技優領航專班
課程科目表

Kun Shan University Day Program Department of Information Engineering Elite Pilot Program for Smart Mechanical Skills

Four-Year Technical Program Course Schedule

課程名稱(講授時數－實習時數－學分數)
Course Title (Lecture Hours – Lab Hours – Credits)

111.11.10 系教學與課程規劃委員會議通過
111.11.14 院課程委員會議通過
112.02.13 校課程委員會議通過
112 學年度入學適用
Effective for students admitted in Fall, 2023

第一學年 1st Academic Year		第二學年 2nd Academic Year		第三學年 3rd Academic Year		第四學年 4th Academic Year	
第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester	第二學期 2nd Semester	第一學期 1st Semester	第二學期 2nd Semester
共同必修科目(合計 30 學分) University-wide Required Courses (30 Credits)							
體育分組(一) Physical Education (I) (2-0-2) 中文閱讀與表達（一） Chinese Reading and Expression (I) (2-0-2) 英文聽力與閱讀(一) English Listening and Reading (I) (2-0-2)	體育分組(二) Physical Education (II) (2-0-2) 勞作教育(一) Labor Education(I) (0-1-0.5) 中文閱讀與表達（二） Chinese Reading and Expression (II) (2-0-2) 英文聽力與閱讀(二) English Listening and Reading (II) (2-0-2) 校訂 分類通識 School-designated Liberal Education (2-0-2)	勞作教育(二) Labor Education(II) (0-1-0.5) 校訂 分類通識 School-designated Liberal Education (2-0-2) 院訂 核心通識 **科技英文** College-designated General Core **Scientific and Technological English** (2-0-2) 院訂 分類通識 College-designated Liberal Education (2-0-2)	服務學習與公民教育 Public Service and Civic Education (0-1-0) 校訂 分類通識 School-designated Liberal Education (2-0-2)	生涯發展與職業倫理 Career Development and Work Ethics (2-0-2) 院訂 分類通識 College-designated Liberal Education (2-0-2)	校訂核心通識 --外語選讀*** School-designated General Core *** (2-0-2)		*英文能力檢定 English Proficiency Examination (1-0-1)
6-0-6	8-1-8.5	6-1-6.5	2-1-2	4-0-4	2-0-2		1-0-1
共同選修科目(合計 1 學分) University-wide Elective Courses (1 Credit)							
		外語選讀 (第二外語)*** Foreign Language second language (2-0-2)	外語選讀 (第二外語)*** Foreign Language second language (2-0-2)		外語選讀(職場英文) *** Foreign Language-Workplace English (2-0-2)		跨領域學習護照** Interdisciplinary Learning Passport ** (1-0-1)
專業必修科目(合計 25 學分) Professional Required Courses (Total: 25Credits)							
程式設計(一) Programming Design (I) （1-2-3）	程式設計(二) Programming Design (II) （1-2-3）	資料結構 Data Structure (3-0-3)					程式設計 Programming Design
		資料庫實務 practice of database (1-2-3)	軟體工程 Software Engineering (3-0-3)				軟體開發 Software Development

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			計算機網路實務 Computer Network Practice (1-2-3)		創新智慧生活科技 Innovative Smart Life Technology (2-1-3)			雲端運算與網路技術 Cloud Computing and Network Technology
					*專題製作（一） Special Project (I) （0-2-2）	*專題製作（二） （0-2-2） Special Project (II) 學術倫理* Academic Ethics (0-1-0)	專業認證 Professional certification (0-1-0)	整合性課程 Integrated Courses
1-2-3	1-2-3	4-2-6	1-2-3	3-0-3	2-3-5	0-3-2	0-1-0	
專業選修科目(依每學期實際情況開課，合計 37 學分，含專業養成課程 8 學分) Professional Elective Courses (Offered based on the actual circumstances of each semester, 35 Credits Total, including 8 credits of professional development courses)								
	HTML 5 程式設計 HTML 5 Programming (1-2-3)		伺服網頁程式設計 Server Web Design (1-2-3)					
數位邏輯與實習 Digital Logic and Practice (2-1-3) 微積分 Calculus (3-0-3)	工程數學 Engineering Mathematics (3-0-3)	影像處理概論 Introduction to Image Processing (3-0-3) Linux 作業系統實務 Practice of Linux Operating System (1-2-3)	演算法 Algorithm (3-0-3) 伺服器架設實務 Server Construction Practice (1-2-3)	計算機結構 Computer Architecture (3-0-3) 嵌入式系統與實習 Course outline of Embedded System and Lab (1-2-3)	離散數學 Discrete Mathematics (3-0-3)	專案管理 Project Management (3-0-3)	電子商務實務 Practice of E-Commerce (3-0-3)	計算機數學與理論 Mathematics for Computing and Theory
工程技術概論 Introduction to Engineering Technology (2-0-2)	物聯網於綠能之應用 IOT application in Green Energy (1-2-3)	App 程式設計 App Programming (1-2-3)	微算機原理與實習 Microcomputer principle and practices (1-2-3)	智慧能源與物聯網 Intelligent Energy and IOT (1-2-3)		物聯網實務 Practice of Internet of Things (1-2-3)	多媒體智慧聯網實務 Multimedia Smart Networking Practice (2-1-3)	智慧聯網 Smart Network
資訊工程導論 Outline of Information Technology Concepts (3-0-3)				雲端運算概論 Introduction to Cloud Computing (3-0-3) 大數據分析 Big Data Analysis (1-2-3)			雲端服務導向架構中介軟體 Cloud Service-Oriented Architecture Middleware (3-0-3)	智慧服務 Intellegient Service
基礎滲透測試 Fundamental Penetration Testing (1-0-1)	網路攻防實務 Cyber Offense and Defense Practice (1-0-1) CTF 搶旗大賽實務技術 CTF Flag-Raising Contest Practical Technology (1-0-1))	網站安全 Web Security (2-1-3)	系統安全 System security (3-0-3)	雲端安全 Cloud Security (3-0-3)			安全實務 Safety practice (2-1-3) 物聯網與資訊安全 Internet of Things and Information Security (2-1-3)	資訊安全 Information Security

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基本電學與實習 Fundamental Circuit and the Practice （1-2-3）	可程式邏輯與實習 Programmable Logic and Practice （1-2-3） 人工智慧概論 Introduction to Artificial Intelligence (3-0-3)	電子電路與實習 Electronics and the Practice (1-2-3)		嵌入式微處理器系統 Embedded Microprocessor Systems (1-2-3) 電腦視覺實務 Computer Vision Practice (1-2-3)	影像處理 Image Processing (3-0-3) AOI 檢測技術 實務 AOI testing technology practice (1-2-3) 機器學習 Machine Learning (3-0-3)	資電整合實務 Integration of Information and Electrical Engineering (1-2-3) 人工智慧 Artificial Intelligence (3-0-3) 深度學習實務 Deep Learning Practice (3-0-3)	機器人程式設計 Robotic Programming (1-2-3) 智慧聯網自動光學檢測實務專題 Smart Networked AOI Special Project (1-2-3)	智慧應用 Intelligent Application
		Swift IOS app 開發 Swift iOS Applications Development (3-0-3)	行動裝置 App 專題製作 Mobile Device App Project Creation (3-0-3)	iOS 應用開發 iOS app Development (3-0-3)	行動裝置遊戲開發 Mobile Device Game Development (3-0-3)	行動 APP 專案開發 Mobile APP Project Development (3-0-3)		App 開發 App Development
跨網頁平台設計 Cross-web platform design (3-0-3)	網站開發與安全 Web Development and Security (3-0-3)				D3.js 程式設計 D3.js programming (3-0-3)	JSP 程式設計 JSP Programming (3-0-3)	Ajax 程式設計 Ajax Programming (3-0-3)	網站程式設計 Web Programming
python 程式設計 python Programming （3-0-3）	C#程式設計 C# Programming (1-1-2)						Design Pattern 實務 Design Pattern Practice (1-1-2)	程式設計 Programming Design
視覺設計實務 Visual Design Practice (3-0-3)	網頁視覺設計實務 Web Visual Design Practice (3-0-3)			手機 UI 設計 Mobile UI Design (3-0-3)		資料視覺化 Data Visualization (3-0-3)		互動設計 Interactive Design
						學期校外實習(一) Semester Internship (I) (0-*-9) 學年校外實習(一) Annual Internship (I) (0-*-9) 暑期校外實習 Summer School Internship (0-*-4)	學期校外實習(二) Semester Internship (II) (0-*-9) 學年校外實習(二) Annual Internship (II) (0-*-9)	企業實習 Industrial Internship

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微服務伺服器實務 Microservice server practice (0-1-1) 專案開發與應用實務 Project Developement and Application Practice (0-1-1) 關聯式資料庫實務 Relational Database Practice (0-1-1)	GTM 實務 Google Tag Manager Practice (0-1-1) 軟體測試實務 Software Testing Practicalities (0-1-1) 人機介面設計專題 Human Machine Interface Design Project (0-1-1)	資電整合應用 Applications of Information and electronic integration (0-1-1) 跨平台程式設計專題 Cross-platform programming languages (0-1-1) 網頁前後端整合實務 Web Integration Practices on Front-End and Back-End (0-1-1)	M2M 聯網實務 Practice of Machine to Machine (0-1-1) 爬蟲程式應用實務 Application of python in data crawling (0-1-1) 物聯網巨量資料實務 IOT BigData Practice (0-1-1)	微電腦視覺實務 Microcomputer Vision and Practice (0-1-1) 網路串流技術應用 The Application of Network Streaming Technology (0-1-1) 機器學習實務 Machine Learning Practice (0-1-1)	微控制器資訊感測 The sensing of microcontroller (0-1-1) 行動影音技術專題 Audiovisual Technology in Mobile Devices (0-1-1) iOS 行動應用實務 iOS Mobile App Practices (0-1-1)	機器手臂應用 Robot Arm Applications (0-1-1) CTF 資安實戰 CTF and Security (0-1-1) AI 生醫感測實務 AI Biomedical Sensor and Measurement (0-1-1)	感測器應用實務 Sensor Applications (0-1-1) 機器學習與資訊安全實務 Machine learning and Information Security Practice (0-1-1) 物聯網應用實務 IoT Appilication Practice (0-1-1)	專業養成課程 Professional Development Courses

技優生推薦選修之三創與跨領域課程 Recommended CIE and Interdisciplinary Courses for Skill-Based Elite Students

※三創選修課程(合計 18 學分) Innovation, Creativity, and Entrepreneurship Courses (Total: 18 Credits)

創意程式設計-C-E Innovative Programming (2-0-2) 創意媒體與生活 Introduction to Creative Media and Life (1-0-1)	創意故事 Creative Writing (2-0-2) 創意管理 Creativity Management (3-0-3)	創意電子控制工具使用入門-E Getting Started with Creative Eletronic Control Tools (2-0-2) 創意媒體英文表達-院核心 English for Creative Media (2-0-2) 創意文案撰寫 Innovative Copywriting (1-2-3)	創意行銷 Creative Marketing (2-0-2) 創意開發 Creative Development (2-0-2) 電腦輔助創新設計 Innovative Computer Aided Design (1-2-3)	綠色科技與創新-C Green Technology and Innovation (2-0-2) 創意思考與方法 Creativity and Change (2-0-2) 廣告創意與文案 Advertising Creativity and Copywriting (1-1-2)	創新事業導論-B Introduction to Entrepreneurship (2-0-2) 創新智慧生活科技 Innovative Wisdom Life Science and Technology (2-1-3)	創意思考訓練 Original Thinking Training (2-0-2) 創意與專利 Ideas and patents (3-0-3) 創意與創業 Creativity an entrepreneursh (3-0-3)	就業與創業技能輔導 Employment and entrepreneurial skills counseling (3-0-3) 環保與創意 Environmental Protection and Creation (3-0-3) 創業與經營實務 Undertaking to Start and Business Management Practice (2-1-3)

※跨領域選修課程(合計 18 學分) Interdisciplinary Elective Courses (Total: 18 Credits)

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綠能科技與實作 Green technology and practices (2-1-3) 太陽能工程概論 Introduction To Solar Energy Engineering (3-0-3) 燃料電池應用技術 Fuel Cell Application Technology (3-0-3) 電子商務 Electronic Commerce (3-0-3)	工程材料學 Engineering Materials (3-0-3) 工程技術概論 Introduction to Engineering Technology (2-0-2) 遊戲企劃 Game planning （3-0-3）	智慧車輛控制系統 Intelligent Vehicle Control Systems (3-0-3) 智慧製造技術 Technology of Smart Manufacturing (3-0-3) 多媒體製作 Multimedia Production (3-0-3) 遊戲行為分析 Game behavior analysis (2-0-2)	物聯網與大數據於智慧製造應用 IOT and Big Data analysis applied in Intelligent Manufacturing (3-0-3) 工業 4.0 特色技術 Industry 4.0 featured technologies (3-0-3) 智慧商務應用 Business Intelligence Application (3-0-3) 智慧商圈管理 Introduction to Intelligent Business Management (3-0-3)	新能源與人文社會社會發展 New energy and humanistic and social development (2-0-2) 綠色能源概論 Introduction to Green Energy (3-0-3) 材料科學導論 Elements of Materials Science (3-0-3) 綠建築與智慧建築 Green Building and intelligence Building (3-0-3) 行動遊戲製作 mobile game production (2-1-3)	儲能系統概論 Introduction to Energy Storage Systems (3-0-3) 綠色能源與產業商機 Introduction to Green Energy and Industry (3-0-3) 企業節能減碳概論 Introduction of Energy Saving and Carbon Reduction (3-0-3) 環境規劃與管理 Environmental Planning and Management (3-0-3) 能源應用 Energy applications (3-0-3) 遊戲行銷營運實務 Game Marketing Operation Practice (0-2-2)	環工生質能源 Biomass energy in environmental engineering (3-0-3) 溫室氣體管理 Management of Green House Gases (3-0-3) 光電材料 Optoelectronic Materials (3-0-3) 節能材料概論 Introduction to Energy Conservation (3-0-3) 太陽光電能發電系統 Photovoltaic Energy Generation System (3-0-3)	物聯網於綠能之應用 IOT application in Green Energy (3-0-3) 智慧能源與物聯網 Intelligence Energy and IOT (3-0-3) 物聯網與綠能概論 Introduction of IOT applied in green energy (2-0-2) 物聯網與綠能應用概論 Introduction to the Internet of Things and Green Energy Applications (3-0-3) 物聯網與綠能應用實務 Practices to the Internet of Things and Green Energy Applications (3-0-3)

最低畢業學分：128 學分 Minimum for Total Credits: 128 Credits

註：

1. **本系畢業學分為 128 學分，必修 55 學分、選修 37 學分、三創課程 18 學分、跨域選修 18 學分。**
   The total graduation requirement for this department is 128 credits, comprising 55 required credits, 37 elective credits, 18 credits in Innovation, Creativity, and Entrepreneurship (CIE) courses, and 18 interdisciplinary elective credits.
2. 選修科目，本系得視實際需求修訂之。
   The elective courses may be revised by the department as needed.
3. 專業必修課程第一次修習於本系修課，再次修習時須事先徵得系之同意後可在他系修習同性質之課程。
   Professional required courses must be taken within this department for the first attempt. If retaken, approval from the department is required before enrolling in an equivalent course offered by another department.
4. 上列**專業選修**科目，自行網路選課，低年級除特殊情況經系主任同意者外不得修習高年級課程。
   The above professional elective courses should be selected online by the students themselves. Lower-year students are not allowed to take upper-year courses unless special permission is granted by the department chair.
5. **學生除選修本系專業選修課程外，可多選修產業學程、跨院系跨領域學程及課程等，以符合就業市場之需求，增加就業力。**
   In addition to taking the department's professional elective courses, students are encouraged to take additional industry-oriented, interdisciplinary, or cross-college programs and courses to meet job market demands and enhance their employability
6. **系專業必修科目「專業認證」，需於在本校就讀期間取得本系所認定之乙級以上（含）專業證照 1 張、取得參加全國性資訊相關競賽證明乙份、通過系專題展及繳交專題報告，始得通過。**
   For the departmental required course **"Professional Certification"**, students must, during their period of study at the university, obtain **at least one professional certificate at Class B or higher** as recognized by the department, **one proof of participation in a national information-related competition**, and **successfully complete the departmental project exhibition and submit a project**

report in order to pass the course.